

Ethan Lin Wong

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EDUCATION

University of California, Irvine

Irvine, CA

Master of Data Science

Expected Dec. 2026

- **Relevant Coursework:** Artificial Intelligence, Machine Learning and Data Mining, Generative Models (in progress), Bayesian Data Analysis, Statistical Methods I & II, Algorithms for Data Science, Databases

University of California, Irvine

Irvine, CA

Bachelor of Science in Data Science

June 2025

EXPERIENCE

Emission Modeling Intern

Sept. 2026 – Present

California Air Resources Board, Air Quality Planning and Science Division

Riverside, CA

- Documenting an internal VBA-based modeling system to support a potential migration to Python
- Building UML and database diagrams, with overviews for leadership and variable-level detail for engineers

Research Assistant

July 2026 – Present

Fortin Lab, University of California, Irvine

Irvine, CA

- Built interpretable models (logistic regression on frequency-band features) that predict from rat hippocampal LFP recordings whether an odor sequence was in order, reaching 73% mean balanced accuracy across 5 rats
- Decoded which of 5 odors a rat was sniffing at 52% mean balanced accuracy against a 20% chance level
- Developed a sliding-window analysis of the GLM coefficients showing when each frequency band drives predictions
- Presented findings to lab members and faculty using a slide deck and notebook built for non-specialists, with weekly progress updates to a postdoctoral supervisor

Statistics Tutor

Sept. 2022 – Aug. 2023

Pasadena City College

Pasadena, CA

- Delivered 500+ hours of remote one-on-one instruction in introductory and intermediate statistics, clarifying probability, numerical analysis, and hypothesis testing

PROJECTS

Stock Directional Forecasting

Feb. 2026 – Mar. 2026

- Built a classifier predicting next-day stock direction with 57.3% precision, using 20 engineered features from 10 years of daily price data
- Beat a random-walk baseline by 14.6% in precision using a Random Forest and Logistic Regression ensemble

Rubik's Cube Reinforcement Learning Solver

Nov. 2025 – Dec. 2025

- Built a hybrid solver combining hill-climbing with random exploration, reaching a 26.9% success rate on short scrambles in a space of 43 quintillion states
- Improved solve reliability 8x over a reinforcement learning baseline by designing a reward function based on correctly placed stickers

Wildfire Smoke Dispersion Modeling | 2025 ICS Project Expo

Jan. 2025 – May 2025

- Led a team of four, coordinating weekly meetings and communication with faculty
- Built the temporal component of the geospatial pipeline, aligning NOAA fire and smoke events in time and aggregating 12-hour post-fire weather features from Open-Meteo
- Contributed to an XGBoost model that more than doubled the R-squared of linear baselines (0.45 vs. 0.20), with SHAP analysis showing fire size and wind speed as top predictors

TECHNICAL SKILLS

Machine Learning: Deep Neural Networks (CNN, RNN), Reinforcement Learning, Supervised & Unsupervised Learning, Logistic Regression (GLM), Random Forest, XGBoost, Scikit-learn, Model Explainability (SHAP)

Programming: Python, C++, R, SQL (PostgreSQL/PostGIS, MySQL), Bash

Libraries: PyTorch, TensorFlow, Keras, NumPy, Pandas, GeoPandas, SciPy, Matplotlib, Seaborn, Plotly

Data & Statistics: Signal Processing (FFT), Cross-Validation, Time-Series Analysis, Bayesian Inference, A/B Testing

Tools: Docker, Git/GitHub, Tableau, Jupyter Notebooks, LaTeX